

Monte Carlo Cone

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Distribution of possible outcomes

PROFESSIONAL DEFINITION

The Monte Carlo Cone visualises the distribution of possible future outcomes generated by simulating many random scenarios, replacing a single point forecast with a fan of probabilities.

WHO DEVELOPED IT

Based on the Monte Carlo method developed by Stanislaw Ulam and John von Neumann, widely applied in finance.

WHY IT WAS DEVELOPED

It was developed to represent uncertainty honestly, showing the range and likelihood of outcomes rather than a misleading single estimate.

FIGURE 1 · MONTE CARLO CONE

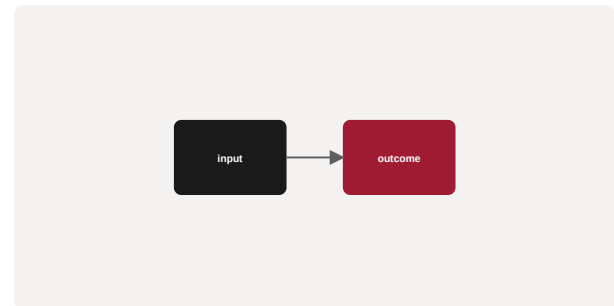


Figure 1. A schematic of the Monte Carlo Cone as applied in BARBM312 (illustrative).

HOW IT IS USED

Many simulated paths are generated from assumed distributions; the resulting cone shows the spread of plausible outcomes for planning.

WHEN IT IS USED

When modelling uncertain investment and financial outcomes.

BY WHOM IT IS USED

Analysts, risk managers and planners.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It communicates uncertainty as a distribution, countering false precision in forecasts — including AI-generated ones.

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SOURCES (HARVARD REFERENCING STYLE)

Metropolis, N. and Ulam, S. (1949) 'The Monte Carlo method', Journal of the American Statistical Association, 44(247).

Glasserman, P. (2003) Monte Carlo Methods in Financial Engineering. New York: Springer.